Shadi Rahimfallah

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PROFILE

Economics Ph.D. candidate at York University (expected 2027) with experience in economic modelling, econometrics, machine learning, and financial and market analysis. Research in firm investment, financial frictions, and technology adoption; electricity market design and market power; and Treasury bond returns. Industry and consulting work in pricing, forecasting, budgeting, and business cases, with a focus on large datasets and turning analysis into decisions for CEOs, shareholders, and client teams. Programming skills in Python, Stata, MATLAB, R, and SQL. Authorized to work in Canada.

EDUCATION

Ph.D. in Economics, York University, Toronto — 2022 – 2027 (expected)

M.Sc. in Energy Economics, University of Tehran, Tehran — 2016 – 2019
Thesis: "Shale Gas and the Export Competitiveness of Iran's Petrochemical Industry"

B.Sc. in Theoretical Economics, Shahid Beheshti University, Tehran — 2012 – 2016

RESEARCH PROJECTS

Growth, Productivity, and Macro-Finance

"Financial Frictions, Used Capital, and Misallocation", *Job Market Paper* [PDF] [SSRN] [Slides] [Code]
Under review, *Journal of Economic Dynamics and Control*
General-equilibrium model with new and used capital and borrowing constraints, solved by dynamic programming; firm fixed-effects regressions on Vietnamese data for calibration: used capital cuts productivity losses by 39%.
➤ Presented at: CEA Annual Meetings (2025, 2026); Macro Development Group, York University (2025, 2026)

"Second-Hand Machines, Firm Capability, and Technology Adoption", draft in progress
Vintage-choice model and firm surveys (15 countries): capability-bound vs. cost-bound users of old machines.

"News, the Economy, and Bond Returns", with Alireza Naeiji, work in progress
Neural-network forecasting: how macroeconomic data and news text predict future Treasury excess returns.

Energy and Electricity Markets

"Electricity Capacity Market", with Jeremy Lin (Johns Hopkins; formerly PJM), work in progress
Large-load market design: how fast-growing data-centre load affects PJM's capacity market, and an auction to serve it.

"Geoeconomics of Global Energy Transformation", with Farkhondeh Jabalameli,
Journal of World Sociopolitical Studies 7(1): 55–87, 2023 [DOI]
ARDL, error-correction, and causality tests: long-run oil pass-through to U.S. polyethylene prices.

Canadian Economy: Public Data Reports

"Canada's Exporters and the U.S. Market, 2000–2025", Statistics Canada data [Report]
85% of goods exporters sell to the U.S. and 65% only there (2025; 56% in 2010); small firms are most dependent.

"How Canadian Firms Work: Evidence from the 2024 Enterprise Survey", survey-weighted [Report]
Obstacles, innovation, finance, trade, and green practices of Canadian firms by size, sector, and region.

PROFESSIONAL EXPERIENCE

Research

Research Assistant, York University — 2023 – 2025
Responsibilities: *Impact evaluation of a World Bank cluster-randomized land-titling trial (Philippines), rebuilt from raw survey data: balance and attrition checks, ANCOVA intent-to-treat effects with clustered standard errors, and tests of consumption insurance; built a World Bank–FAO country–year panel on trade risk and food security.*
Tools & Technologies: *Python, Stata.*

Research Assistant, University of Tehran 2018

Responsibilities: *Studied Iran's day-ahead electricity market design and regulation and analyzed generator bidding data for market power; compared administered and market gas prices; estimated two-way fixed-effects models of how supply shocks and export credit shape Iran's export diversification.*
Tools & Technologies: *MATLAB, Stata, Python.*

Industry and Consulting

Senior Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis 2019 – 2021

Responsibilities: *Led financial and market analysis for the CEOs and shareholders of a four-company diagnostics group; built business cases for new tests, from market sizing and cost per test through pricing; built demand, price, and cost forecasts for budgeting and business planning; led the case for COVID-19 PCR testing on existing equipment, which became a material contributor to revenue.*
Tools & Technologies: *Stata, SQL, Tableau.*

Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis 2016 – 2019

Responsibilities: *Built demand, pricing, and market-sizing models from sales data; tracked competitor prices and market shares; prepared the budgets and financial reports for the laboratory and three affiliated companies.*
Tools & Technologies: *Excel, SQL, Tableau.*

Economic Analyst, Batab Kara, Economic and Investment Consulting Summer 2017

Responsibilities: *Wrote and presented economic and investment reports for client CEOs and strategy teams; built a house-price forecasting model in MATLAB whose scenarios clients used to time investments; assessed an exporting client's trade risk, including exposure to tariffs and export-market concentration.*
Tools & Technologies: *MATLAB, Stata.*

SKILLS

Programming: Stata, Python, MATLAB, Excel (advanced); R, SQL, Git, LaTeX

Economics: General-equilibrium modelling; DCF/NPV and business cases; forecasting

Econometrics: RCTs and ANCOVA; difference-in-differences; panel data; time series; neural networks

Energy & Competition: Capacity markets (Net CONE, VRR); plant dispatch; market power; market shares

Languages: English (fluent), Persian (native)

TEACHING, LEADERSHIP, AND SERVICE

Teaching Assistant, *York University* 2022 – Present
Project Appraisal, Development Policy, Micro and Macro, Mathematics, Labour

Volunteer Instructor, Graduate Quantitative Skills Initiative, *University of Tehran* 2018 – 2020
Collaborated with a team to provide quantitative and computational training to graduate students

Session Chair, *59th CEA Meetings* (2025); Secretary, *York Economics Graduate Student Assoc.* (2024)

GRANTS & AWARDS

LA&PS Dissertation Fieldwork Fellowship, *York University* 2025

MacroFinance Summer School, full scholarship, *Princeton University* 2025

Computational Methods for Macroeconomics, full scholarship, *Econometric Society* 2023

Decision-Making under Risk, full scholarship, *University of Würzburg* 2021

Full Ph.D. Scholarship, *York*; M.Sc. and B.Sc. Full Scholarships (top 1% nationally), *Iran* 2022, 2016, 2012

REFERENCES

Tasso Adamopoulos (Supervisor), *Professor of Economics, York University* aadamo@yorku.ca

Chaoran Chen (Supervisor), *Associate Professor of Economics, York University* chenecon@yorku.ca

Jeremy Lin (Research Collaborator), *Lecturer, Johns Hopkins University; formerly PJM* jlin137@jhu.edu

Yanran Guo (Committee Member), *Assistant Professor of Economics, York University* yrguo@yorku.ca